# Welcome!

PM 566: Introduction to Health Data Science

# Instructor

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- Office hours: Wednesday 11am, SSB 202V

# Brightspace + Website

Brightspace for announcements and grading:

https://brightspace.usc.edu

Official class website, containing syllabus, reading materials, slides, labs, and assignments:

https://uscbiostats.github.io/PM566/

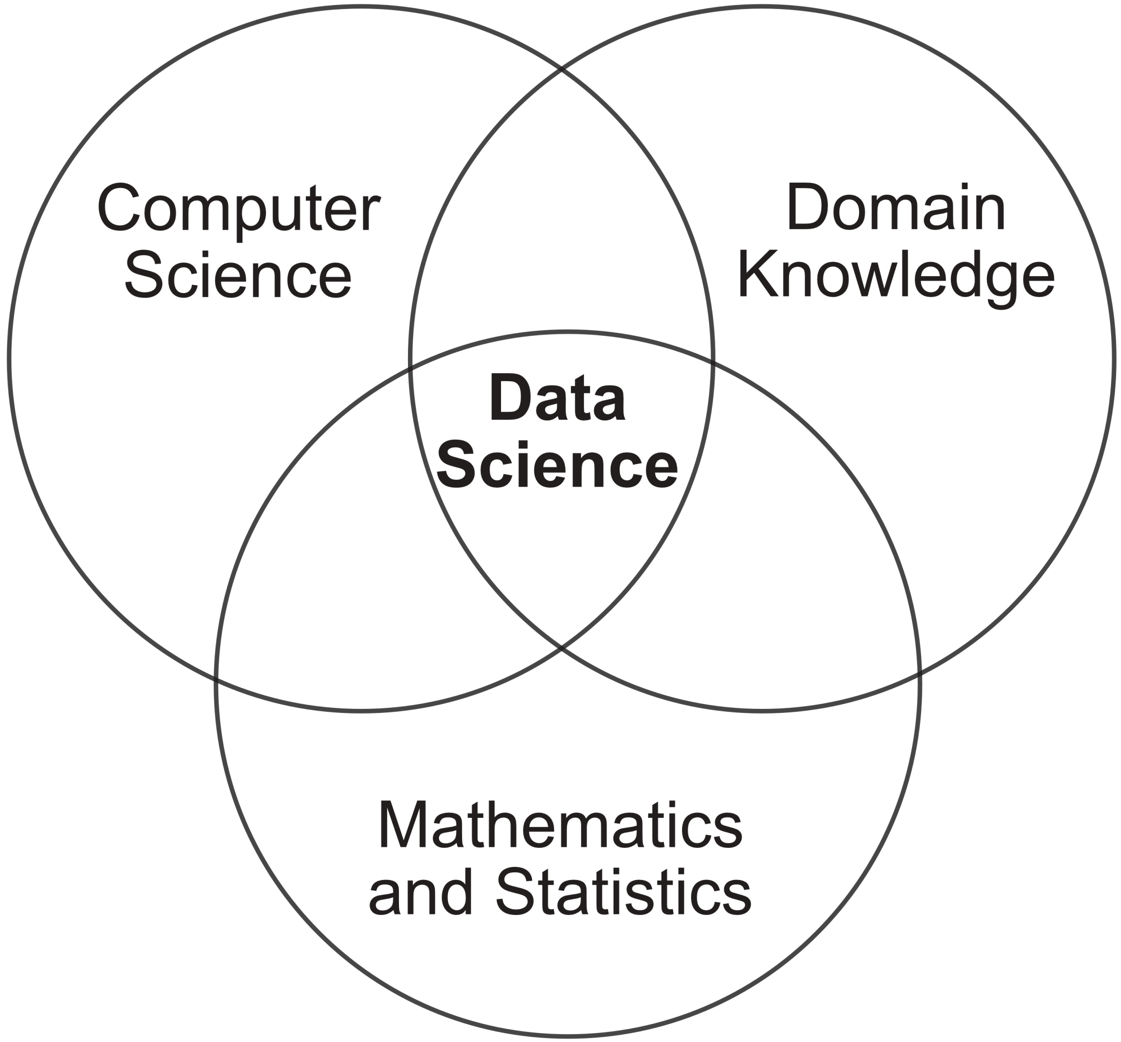
# USC Software Development Help Page

https://uscbiostats.github.io/software-dev-site/

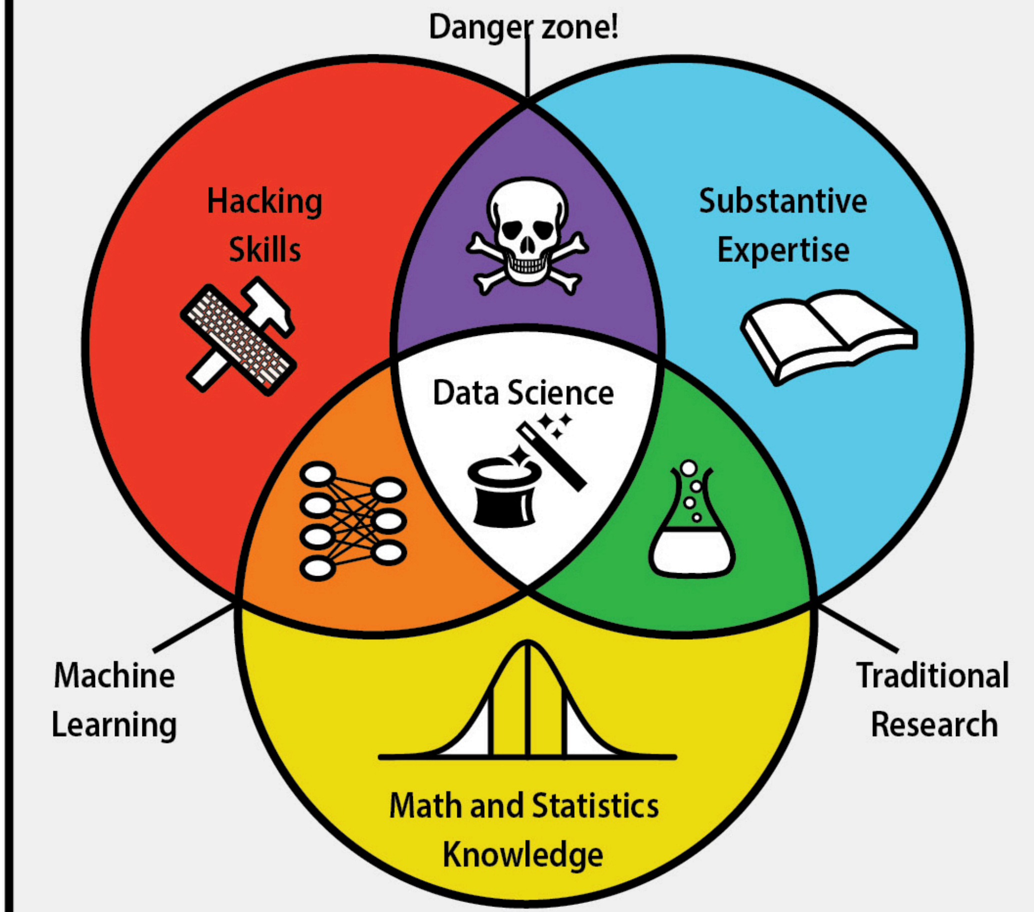
collection of knowledge about

- computing
- standards
- research practices

# What is data science?


Computer Science
Domain Knowledge
Data Science
Mathematics and Statistics

# DATA SCIENCE SKILLSET




Data science, due to its interdisciplinary nature, requires an intersection of abilities: **hacking skills**, **math and statistics knowledge**, and **substantive expertise** in a field of science.

**Hacking skills** are necessary for working with massive amounts of electronic data that must be acquired, cleaned, and manipulated.

**Math and statistics knowledge** allows a data scientist to choose appropriate methods and tools in order to extract insight from data.

**Substantive expertise** in a scientific field is crucial for generating motivating questions and hypotheses and interpreting results.

**Traditional research** lies at the intersection of knowledge of math and statistics with substantive expertise in a scientific field.

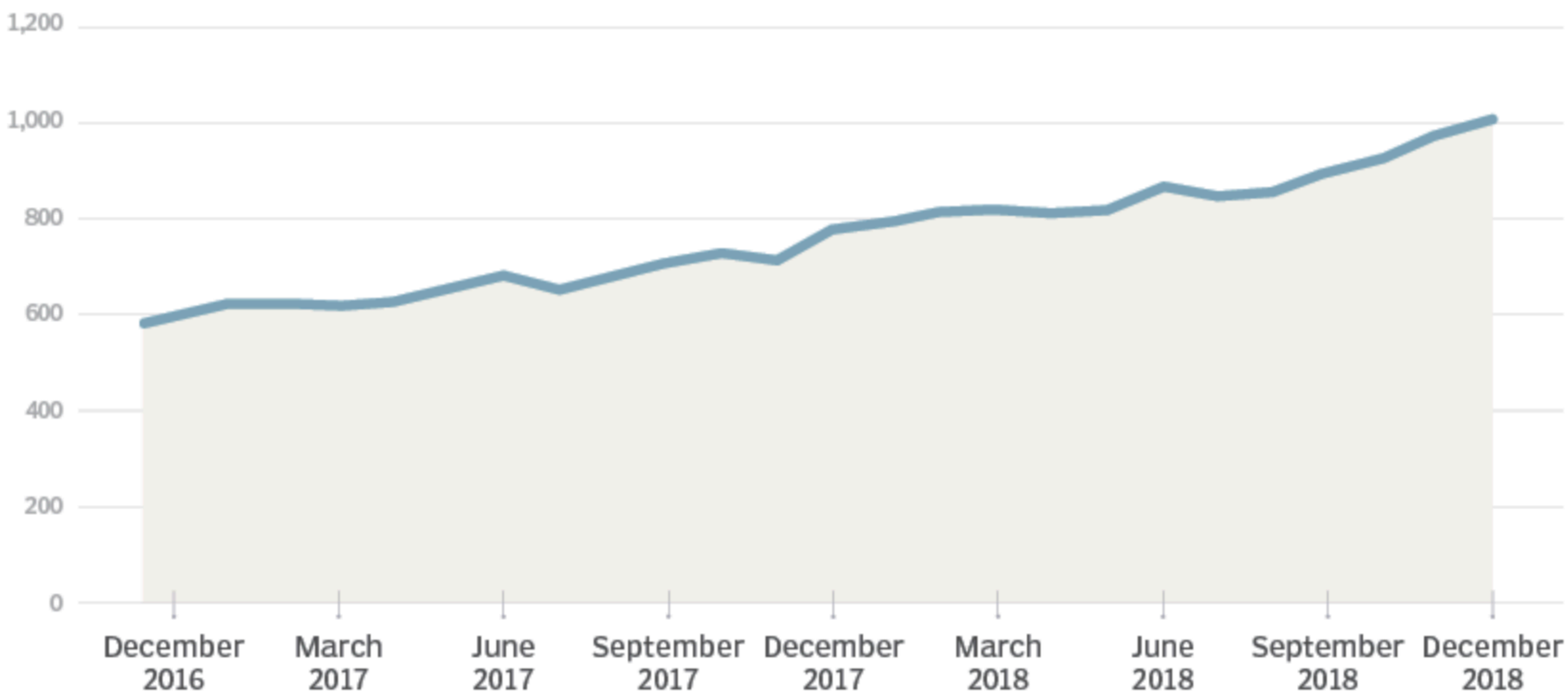
**Machine learning** stems from combining hacking skills with math and statistics knowledge, but does not require scientific motivation.

**Danger zone!** Hacking skills combined with substantive scientific expertise without rigorous methods can beget incorrect analyses.

- Data science is an exciting discipline that allows you to turn raw data into understanding, insight, and knowledge.

# Data scientists are in high demand

Data scientist job postings, per 1 million postings on Indeed




(Also see here, and here, and here, and here )

# What is this course?

- This course is a introduction to the world of data science with a focus on application in the health sciences.
- The course will teach data science skills that are easily transferable, with examples done in R.
- You can use any language/tool you prefer. But we can only guarantee help if you are using R and RStudio.

# What ISN'T this course?

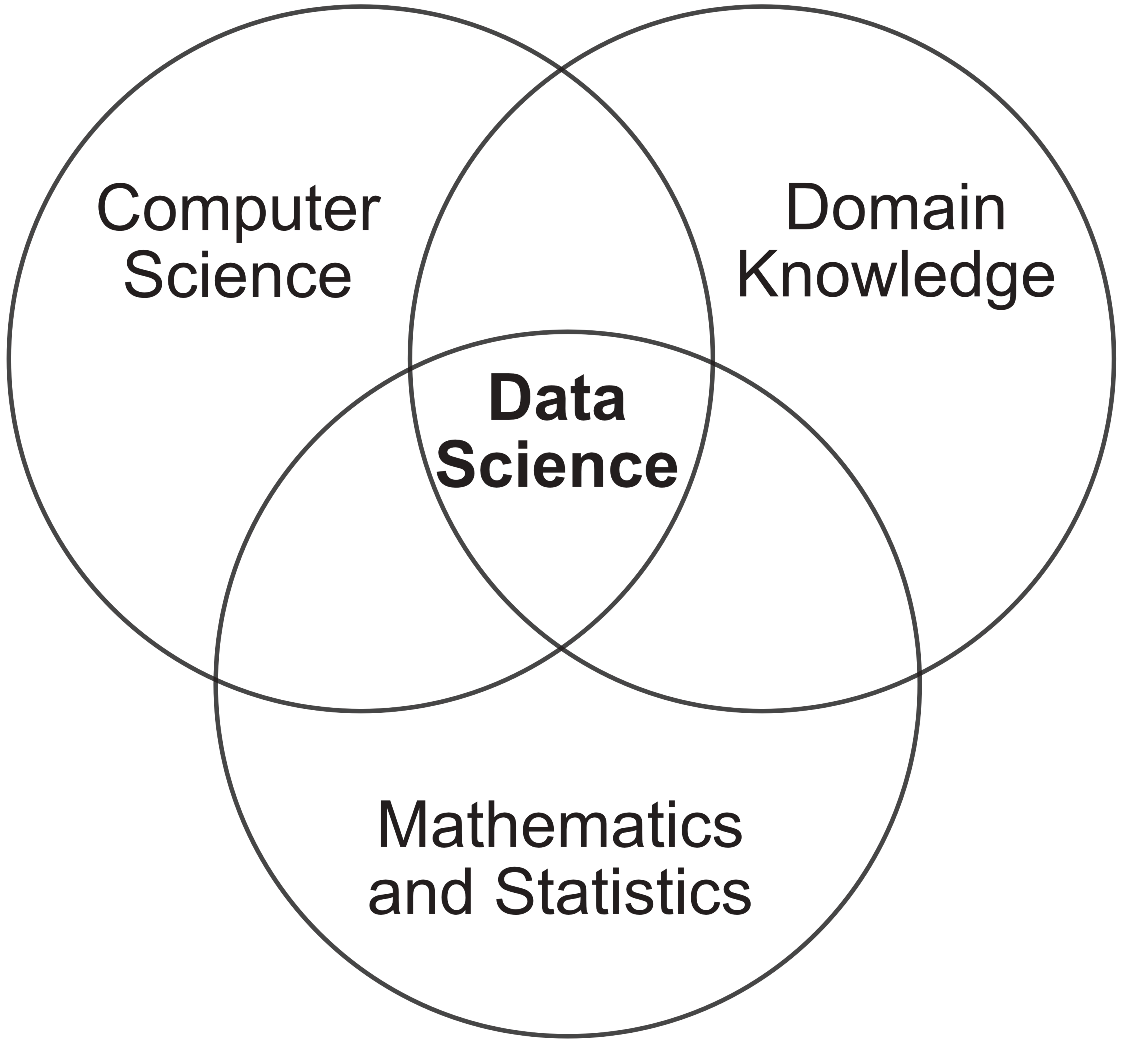
This is not a formal statistics class. You will not be expected to know or use:

- parametric distributions
- hypothesis tests
- p values

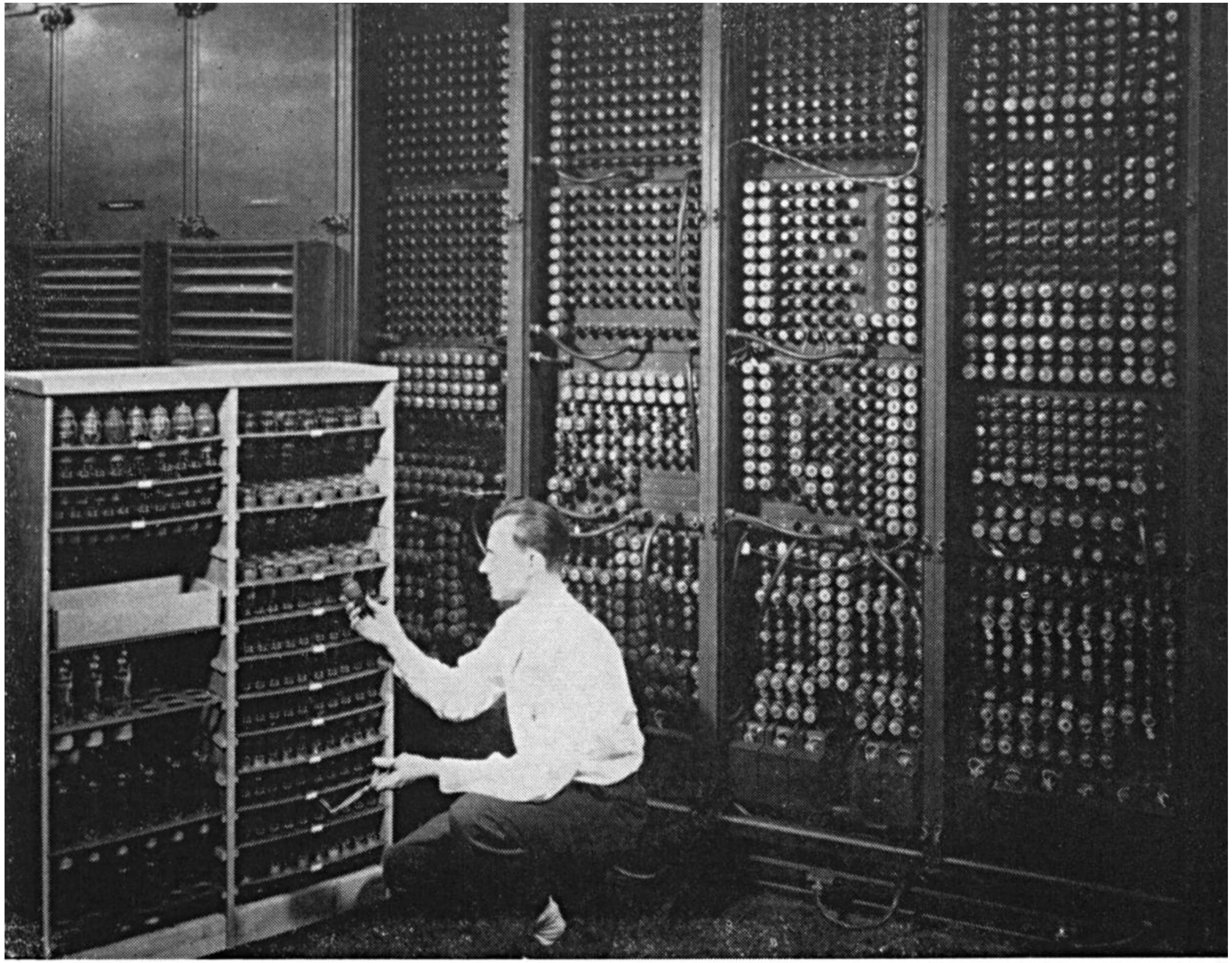
# What ISN'T this course?

Data does not exist in a vacuum. In order to gain new insights from data, you must start with a baseline understanding of the subject. "Domain knowledge" or "subject matter expertise" is *critical*, but it is not the purpose of this class.

This course will focus on applications in Public Health, but the skills you learn will be widely transferable.


Computer
Science
Domain
Knowledge
Data
Science
Mathematics
and Statistics

# What is a computer?



Replacing a bad tube meant checking among ENIAC's 19,000 possibilities.

# File Structure

/ "root"

- /bin "essential user command binaries"
  - bash
  - cat
  - chmod
  - cp
  - date
  - echo
  - grep
  - gunzip
  - gzip
  - hostname
  - kill
  - less
  - ln
  - ls
  - mkdir
  - more
  - mount
  - mv
  - nano
  - open
  - ping
  - ps
  - pwd
  - rm
  - sh
  - su
  - tar
  - touch
  - umount
  - uname
- /etc "configuration files for the system"
  - crontab
  - cups
  - fonts
  - fstab
  - host.conf
  - hostname
  - hosts
  - hosts.allow
  - hosts.deny
  - init
  - init.d
  - issue
  - machine-id
  - mtab
  - mtools.conf
  - nanorc
  - networks
  - passwd
  - profile
  - protocols
  - resolv.conf
  - rpc
  - securetty
  - services
  - shells
  - timezone
- /sbin "essential system binaries"
  - fdisk
  - fsck
  - getty
  - halt
  - ifconfig
  - init
  - mkfs
  - mkswap
  - reboot
  - route
- /usr "read-only user application support data & binaries"
  - /usr/bin "most user commands"
  - /usr/include "standard include files for 'C' code"
  - /usr/lib "obj, bin, lib files for coding & packages"
  - /usr/local "local software"
    - /usr/local/bin
    - /usr/local/lib
    - /usr/local/man
    - /usr/local/sbin
    - /usr/local/share
  - /usr/share "static data sharable accross all architectures"
    - /usr/share/man "manual pages"
- /var "variable data files"
  - /var/cache "application cache data"
  - /var/lib "data modified as programmes run"
  - /var/lock "lock files to track resources in use"
  - /var/log "log files"
  - /var/opt "variable data for installed packages"
  - /var/spool "tasks waiting to be processed"
    - /var/spool/cron
    - /var/spool/cups
    - /var/spool/mail
  - /var/tmp "temporary files saved between reboots"
- /dev "device files incl. /dev/null"
- /home "user home directories"
- /lib "libraries & kernel modules"
- /mnt "mount files for temporary filesystems"
- /opt "optional software applications"
- /proc "process & kernel information files"
- /root "home dir. for the root user"

# File Structure

Before computers had graphics and mice, there were only text-based interfaces, called command lines, that let you interact with the directories and files on the computer.

The modern "Desktop" is actually just a directory on your computer!

- MacOS/Linux: `/Users/<username>/Desktop`
- Windows: `C:\Users\<username>\Desktop`

The route from the root directory to any specific file or directory is called the "path".

# File Structure

Whenever you run a program on your computer, you are running it in a specific location (directory). If you want to access another file on your computer, you'll need to know the path to that file. Paths can be either relative or absolute.

How to get from my `Desktop` directory to my `Documents` directory via:

- absolute path: `/Users/kstreet/Desktop`
- relative path: `../Desktop/`

# File Structure

Special symbols:

- `.` Current directory
- `..` Parent directory (one step up the hierarchy)
- `~` Home directory

We won't have to use the command line too much in this class, but understanding file paths will be very important!

# CARC

At USC, the Center for Advanced Research Computing (CARC) provides students and faculty with high-performance computing capabilities. The interface for working on CARC is entirely text based, meaning it operates via a command line.

https://www.carc.usc.edu/

# What is R?

> R is a language and environment for statistical computing and graphics: https://r-project.org

Created by statisticians for statisticians.

Over 16,000 packages added to CRAN

# What is RStudio?

RStudio logo

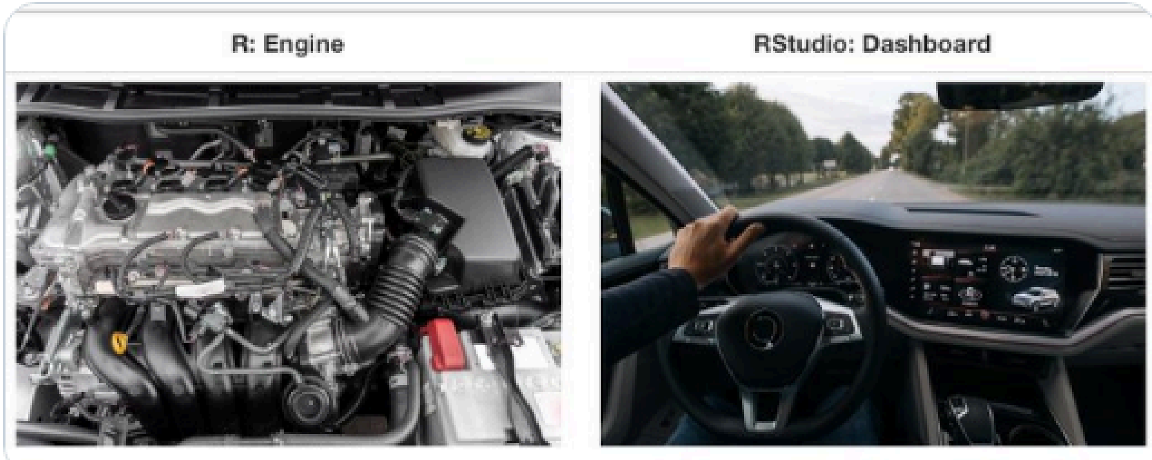
> RStudio is an integrated development environment (IDE) for R: https://www.rstudio.com/products/rstudio/

**ModernDive**
@ModernDive

Now that the school year has begun, let's start at the beginning! Two common questions

First, what's the difference between R & RStudio? Let's use a 🚙 analogy!

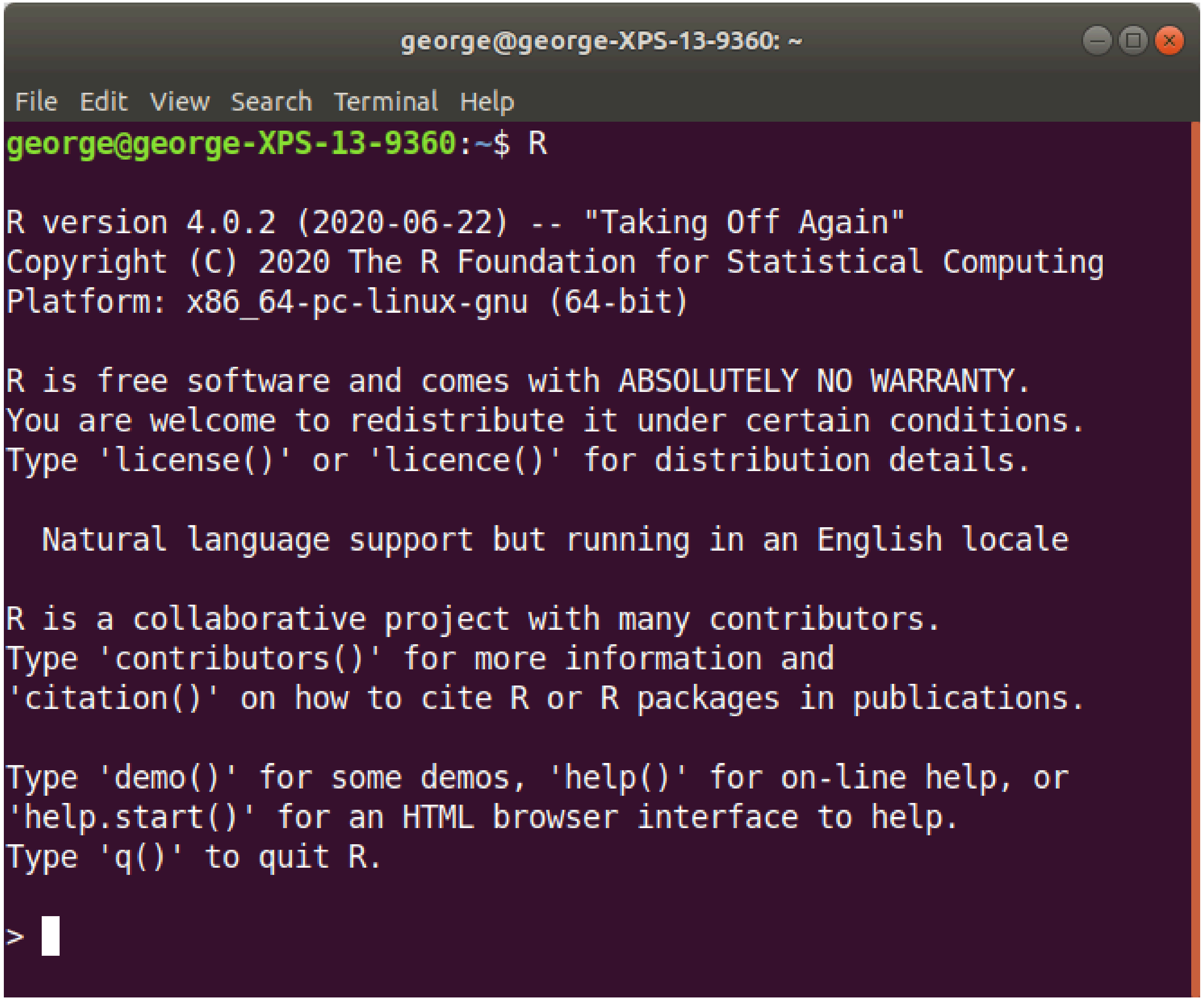
- R is like a car's engine. It does the work!
- RStudio is like a car's dashboard. You interact with R using RStudio!




9:11 AM · Sep 10, 2019

♡ 35 See ModernDive's other Tweets

# R in the terminal

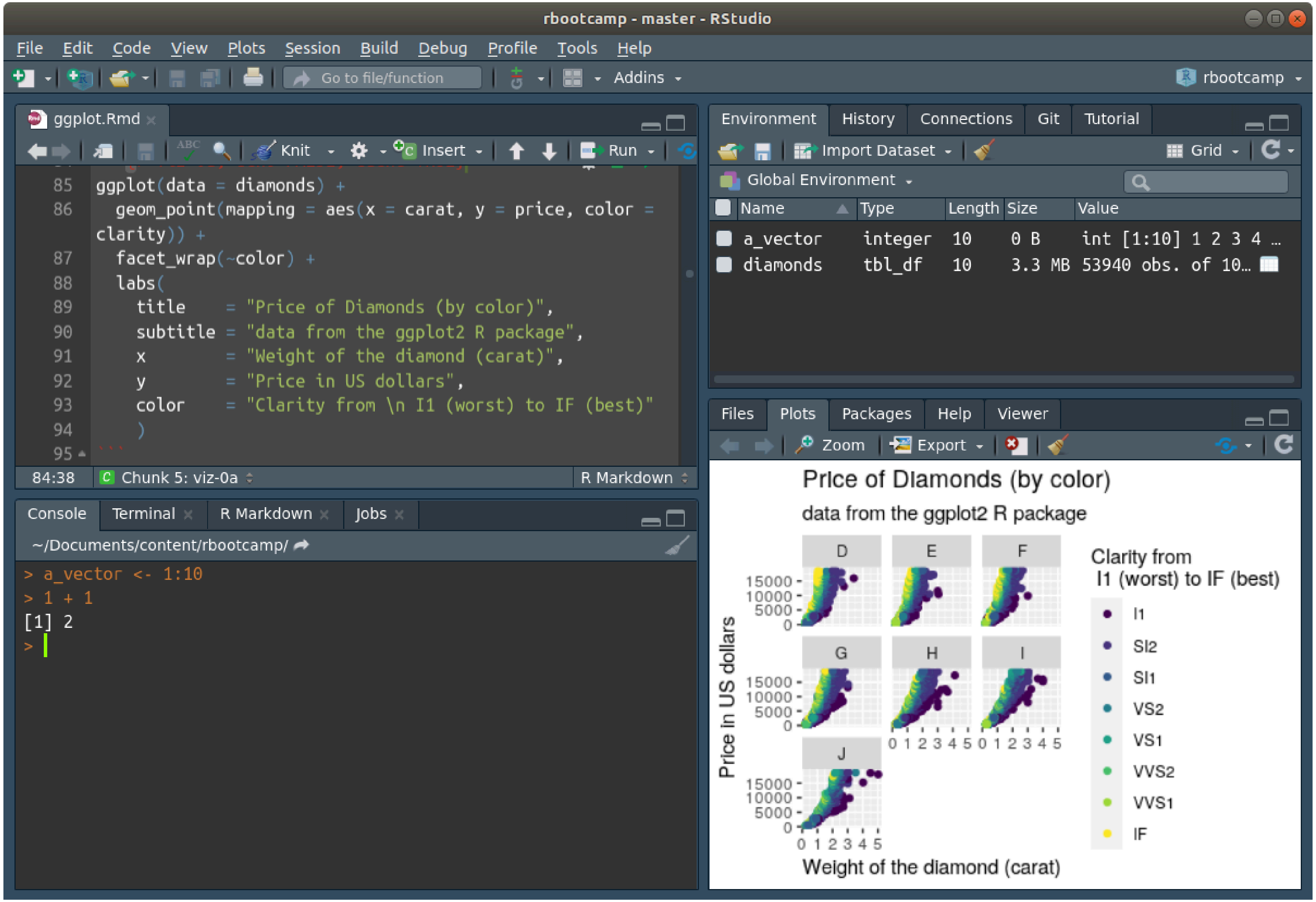


george@george-XPS-13-9360: ~

File Edit View Search Terminal Help

```
george@george-XPS-13-9360:~$ R

R version 4.0.2 (2020-06-22) -- "Taking Off Again"
Copyright (C) 2020 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

  Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

>
```

# R + RStudio



rbootcamp - master - RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
rbootcamp
ggplot.Rmd
Knit
Insert
Run
ggplot(data = diamonds) +
geom_point(mapping = aes(x = carat, y = price, color = clarity)) +
facet_wrap(~color) +
labs(
title = "Price of Diamonds (by color)",
subtitle = "data from the ggplot2 R package",
x = "Weight of the diamond (carat)",
y = "Price in US dollars",
color = "Clarity from \n I1 (worst) to IF (best)"
)
84:38
Chunk 5: viz-0a
R Markdown
Console
Terminal
R Markdown
Jobs
~/Documents/content/rbootcamp/
> a_vector <- 1:10
> 1 + 1
[1] 2
>
Environment
History
Connections
Git
Tutorial
Import Dataset
Grid
Global Environment
Name
Type
Length
Size
Value
a_vector integer 10 0 B int [1:10] 1 2 3 4 …
diamonds tbl_df 10 3.3 MB 53940 obs. of 10…
Files
Plots
Packages
Help
Viewer
Zoom
Export
Price of Diamonds (by color)
data from the ggplot2 R package
Clarity from
I1 (worst) to IF (best)
I1
SI2
SI1
VS2
VS1
VVS2
VVS1
IF
Price in US dollars
Weight of the diamond (carat)

# Break time

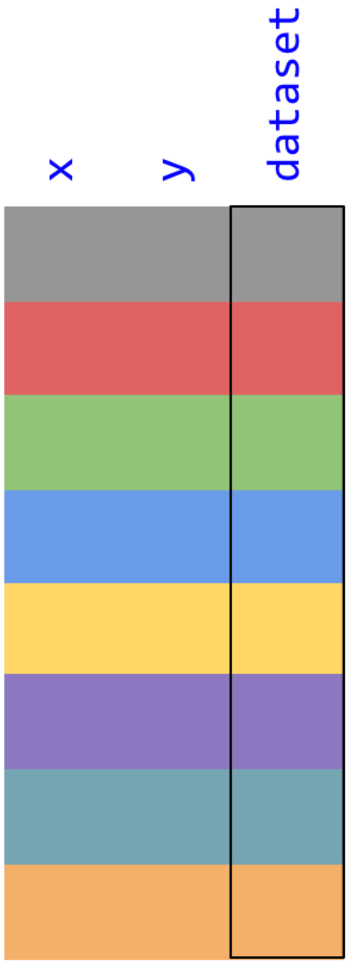
Following break we will run Lab 1

# Dataset

datasaurus_dozen:

# Dataset

`datasaurus_dozen:`



# First Lab

The lab exercises can be found on the Schedule page of the course website:

https://uscbiostats.github.io/PM566/schedule.html